A Brief Statistical Review

ISyE 4031, Week 1 Class 2

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Today: foundations for trustworthy forecasts

By the end of class, you should be able to:

1. distinguish a population, sample, parameter, statistic, and estimand;
2. connect variability to standard errors, intervals, tests, and forecasts;
3. interpret a p-value without turning it into a probability about the null;
4. audit statistical code proposed by an AI agent.

Every formula answers a question under assumptions.

Start with the question, not the software

Unit	What object is observed: a person, day, hospital, state, or model run?
Response	What outcome is measured or forecast?
Population	Which collection of units do we want to understand?
Sample	Which units did we actually observe?
Estimand	What population quantity or future target answers the question?

Forecasting example

Unit: state-week. Response: influenza admissions. Estimand: the predictive distribution for a future state-week under a fixed information cutoff.

Parameters are fixed; statistics vary across samples

Population quantities

- mean μ
- variance σ^2
- correlation ρ
- future outcome Y_{t+h}

Sample quantities

- mean $\bar{y}$
- variance s^2
- correlation r
- forecast $\widehat{Y}_{t+h|t}$

Inference quantifies what sampling leaves uncertain.

Description, inference, and prediction differ

Description	What patterns are present in the observed data?
Inference	What does the sample imply about a population quantity or model parameter?
Prediction	What values are plausible for a new or future observation?

A confidence interval for a mean is not a prediction interval for tomorrow.

Center and spread summarize different features

Center

- mean: sensitive to every value;
- median: robust to extremes;
- mode: most frequent value.

Spread

- range: maximum minus minimum;
- variance and standard deviation;
- interquartile range: robust spread.

Modeling consequence

Similar means do not imply similar uncertainty or similar decisions.

One small sample already contains a modeling decision

For $y = (8, 9, 9, 10, 14)$:

$$\bar{y} = 10, \qquad \tilde{y} = 9, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2 = 5.5, \qquad s \approx 2.35.$$

- The value 14 pulls the mean above the median.
- The standard deviation is in the original units.
- The denominator $n - 1$ estimates a population variance from a sample.

Covariance and correlation preview regression

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \qquad r = \frac{s_{xy}}{s_x s_y}.$$

- Covariance carries the product of the original units.
- Correlation is unitless and lies between -1 and 1.
- Neither quantity establishes causation.
- A strong correlation can still fail as a future forecast.

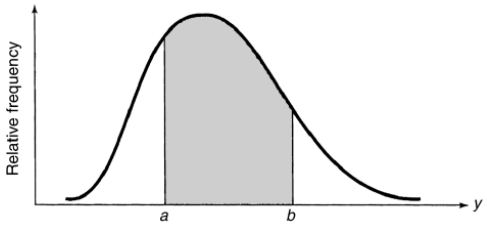
Association is the beginning of a model, not its defense.

Probability models weight possible outcomes

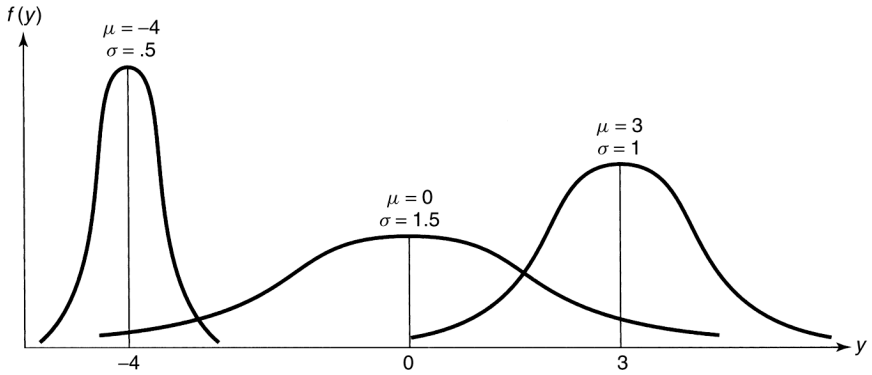
For density f and cumulative distribution F,

- Probability is area under the curve.
- The total area is one.
- Tail areas drive intervals and tests.

- $\mathrm{P}(a \leq Y \leq b) = \int_a^b f(y)dy = F(b) - F(a)$



Mean moves a distribution; spread changes its shape

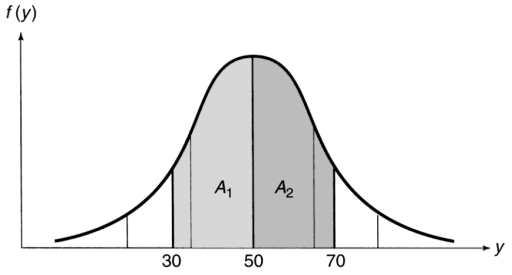


- Changing μ shifts the center.
- Increasing σ spreads probability over a wider range.

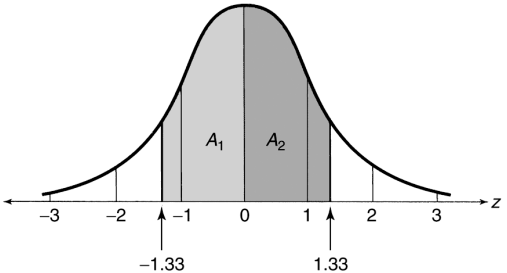
Standardization puts a normal variable on a common scale

If $Y \sim N(\mu, \sigma^2)$, then

$$Z = \frac{Y - \mu}{\sigma} \sim N(0, 1).$$



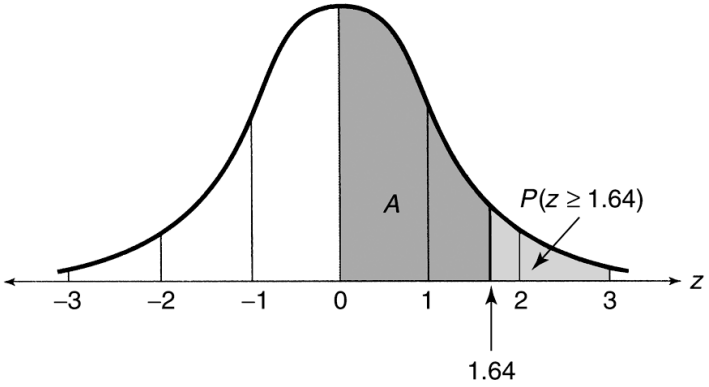
Original scale



Standard-normal scale

The shaded probability is unchanged by standardization.

A tail probability is an area beyond a cutoff



Read the picture

The shaded region is $P(Z \geq 1.64)$, not the height of the curve at 1.64.

Audit habit

Check the tail direction before trusting software output.

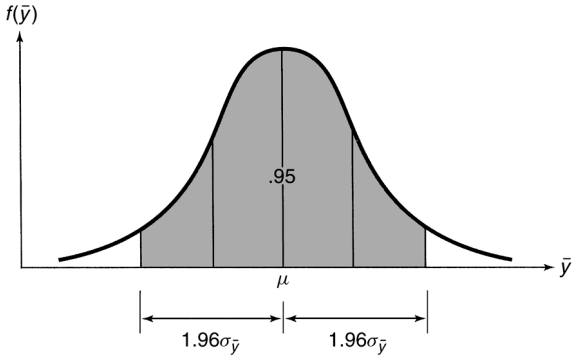
A sampling distribution tracks how an estimator would vary

Repeatedly draw samples of size n and compute $\bar{Y}$:

$$E(\bar{Y}) = \mu, \qquad \mathrm{SE}(\bar{Y}) = \frac{\sigma}{\sqrt{n}}.$$

More data

Observations remain variable, but the sample mean becomes more stable.



The central limit theorem supports approximate inference

If $Y_1, \ldots, Y_n$ are independent and identically distributed with finite mean μ and variance σ^2, then for sufficiently large n,

$$\frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1).$$

- It concerns the distribution of a standardized sample mean.
- It does not say the observed data become normally distributed.
- Dependence, heavy tails, and small samples can make the approximation poor.

Standard deviation is not standard error

Standard deviation s	How dispersed are the observed units?
Standard error $s/\sqrt{n}$	How much would the estimated mean vary across repeated samples?

Common agent error

Dividing by n instead of $\sqrt{n}$ makes uncertainty collapse too quickly.

Confidence intervals target repeated-sampling coverage

With unknown variance and an approximately normal population,

$$\bar{y} \pm t_{1-\alpha/2,\, n-1} \frac{s}{\sqrt{n}}$$

is a $100(1-\alpha)\%$ confidence interval for μ.

Correct interpretation

Across repeated samples, intervals produced by this procedure contain the fixed mean μ about $100(1-\alpha)\%$ of the time.

The interval is not a probability distribution for the already fixed parameter.

Predicting a future observation requires more uncertainty

Under an independent normal model, an interval for a future observation is

$$\bar{y} \pm t_{1-\alpha/2,\, n-1} s \sqrt{1 + \frac{1}{n}}.$$

Mean interval

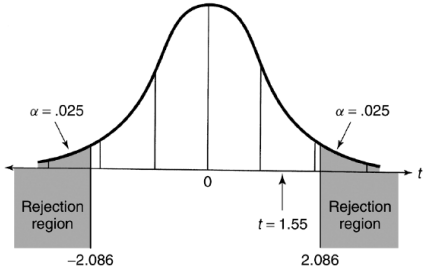
Uncertainty about the unknown population mean.

Prediction interval

Mean uncertainty plus irreducible variation of a new observation.

Hypothesis tests start with a claim and decision rule

1. State H_0, H_a, and assumptions.
2. Choose the statistic and α.
3. Locate the statistic under H_0.
4. Compare its tail area with α.
5. Report the scientific conclusion.



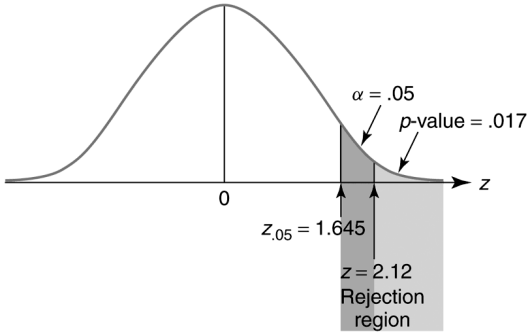
The observed statistic does not enter the rejection region.

A p-value measures compatibility with the null model

Assuming H_0, the p-value is the probability of a statistic at least as extreme as the observed statistic.

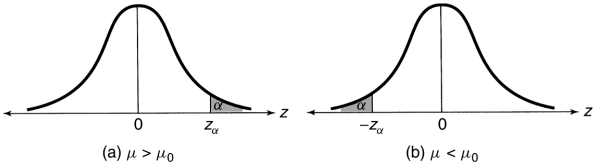
It is **not**:

- $P(H_0 \text{ is true} \mid \text{data})$;
- the probability the result occurred by chance;
- the importance of an effect.



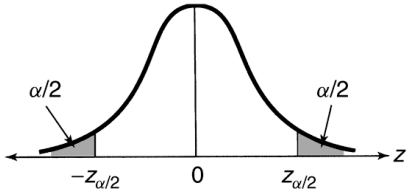
The p-value tail is smaller than the preselected α tail.

The alternative determines the rejection region



(a) $\mu > \mu_0$ (b) $\mu < \mu_0$

One-sided alternatives



(c) $\mu \neq \mu_0$

Two-sided alternative

Choose the direction before looking at the data.

Type I error, Type II error, and power encode tradeoffs

	H_0 true	H_a true
Do not reject H_0	correct	Type II, β
Reject H_0	Type I, α	power, $1-\beta$

- Type I error probability: α.
- Type II error probability: β.
- Power: $1-\beta$.

Power depends on effect size, noise, sample size, the test, and α.

Assumptions are part of the answer

Independence	Observations and errors do not carry unmodeled dependence.
Distribution	The chosen normal, t, or other approximation is adequate.
Stable process	The mechanism does not shift between the sample and target.
Legal information	No future values enter a forecast or its features.

When an assumption fails, change the method or narrow the claim.

AI-native work = baseline + agent + audit

1. **Baseline:** write the question, estimand, assumptions, and simple calculation.
2. **Agent:** request code, tests, or an extension with explicit constraints.
3. **Audit:** inspect formulas, denominators, data cutoffs, packages, and output.
4. **Evidence:** run tests and compare with an independent calculation.
5. **Decision:** accept, revise, or reject the proposal and explain why.

The quality of the audit matters more than the sophistication of the prompt.

Audit this interval before running it

```
x = np.array([8, 9, 9, 10, 14])
mean = x.mean()
sd = x.std()
se = sd / len(x)
interval = stats.norm.interval(.95, loc=mean, scale=se)
```

Find at least three issues

Decide whether the target is the population mean or a future observation, then check the variance denominator, standard error, and reference distribution.

Exit check: name the target and the uncertainty

Before leaving, answer:

1. What is the estimand in a confidence interval for a mean?
2. Why is a prediction interval wider than a confidence interval for the mean?
3. Which line in the buggy code is most dangerous, and how would a test catch it?

Next: simple linear regression turns association into a model.

Optional review: two-sample inference adds a design question

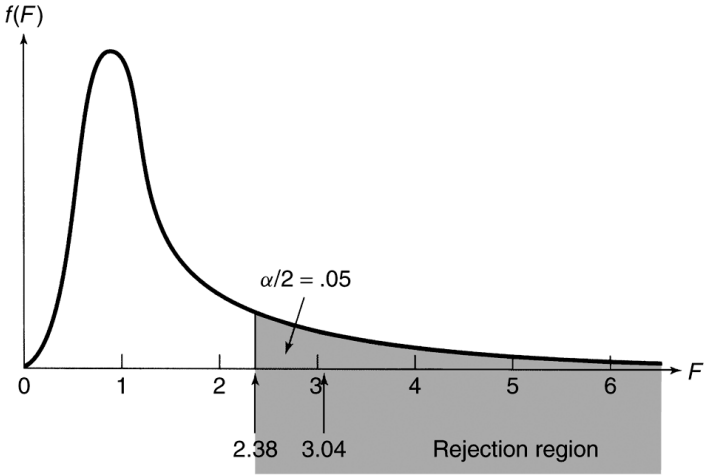
Before comparing two means, ask:

- Are observations independent or paired?
- Is the outcome measured on the same scale?
- Is equal variance defensible?
- Are sampling mechanisms comparable?
- Does the difference answer the estimand?

	Population	
	1	2
Sample size	n_1	n_2
Population mean	μ_1	μ_2
Population variance	σ_1^2	σ_2^2
Sample mean	$\bar{y}_1$	$\bar{y}_2$
Sample variance	s_1^2	s_2^2

Welch's t procedure avoids assuming equal population variances.

Optional: variance ratios and the F distribution



- $F = s_1^2/s_2^2 \geq 0$.
- Degrees of freedom determine the shape.
- Tail placement depends on how the ratio is defined.

Modern practice

Visualize spread and check robustness before relying on a variance test.